

Supporting Information for

Transition metal dichalcogenide nanospheres for high-refractive-index nanophotonics and biomedical theranostics

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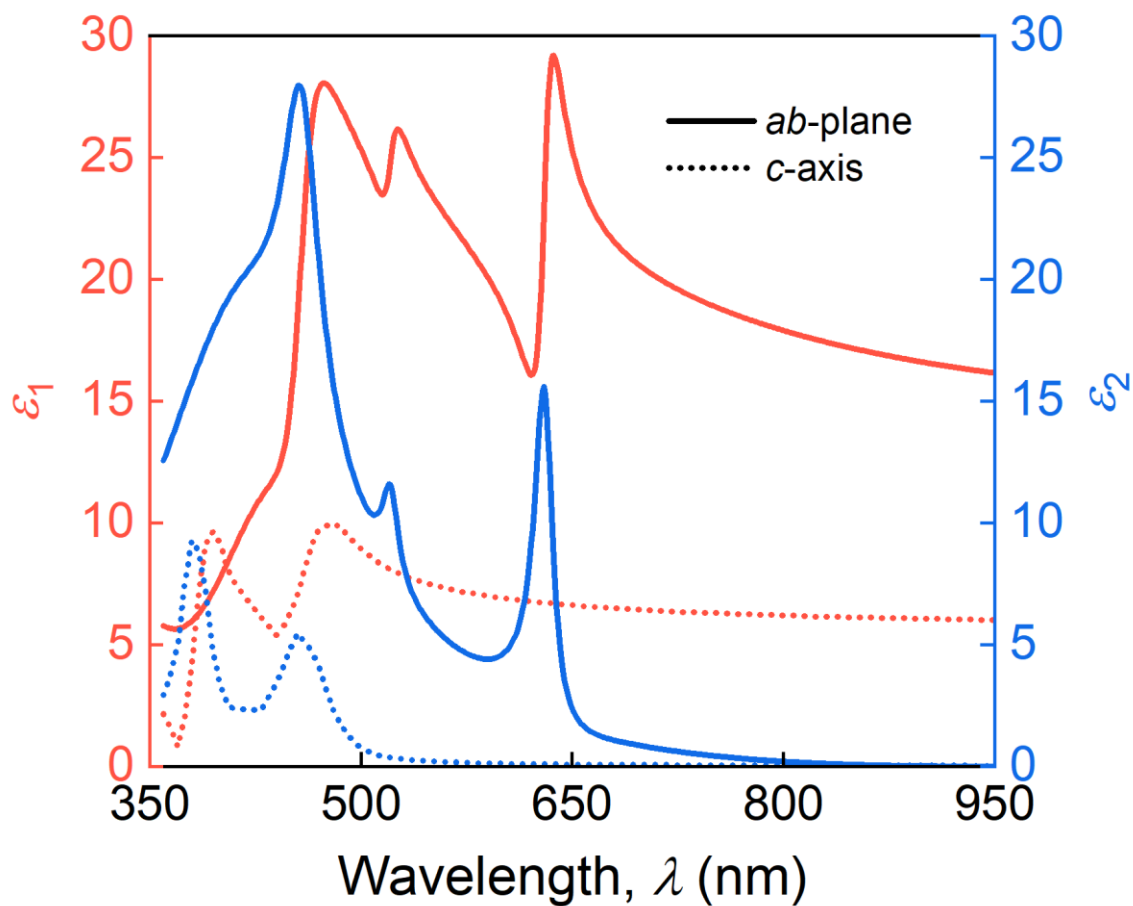
Supporting text

Supplementary Figures 1 to 7

Supplementary Table 1

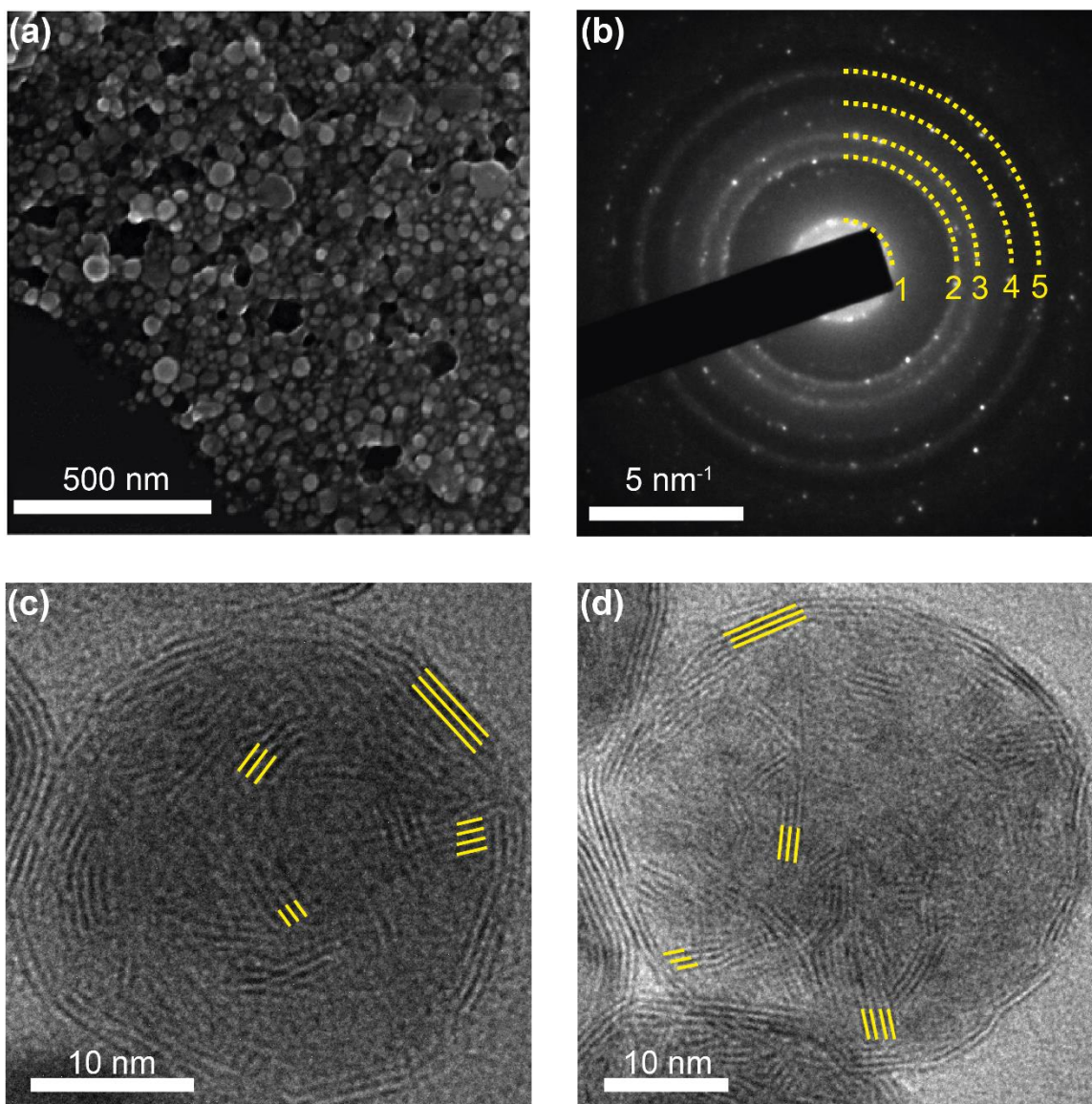
SUPPORTING INFORMATION TEXT

Supplementary Note 1. Dielectric function of WS₂



Supplementary Figure 1. Real (ϵ_1) and imaginary (ϵ_2) parts of the dielectric function along the *ab*-plane and *c*-axis.

Supplementary Note 2. Morphology of WS₂ nanoparticles (NPs)



Supplementary Figure 2. **a** SEM image of laser-ablated WS₂ NPs selected from a colloidal solution at 250 rpm. **b** SAED pattern for laser-ablated WS₂ NPs. TEM images for WS₂ NPs with a diameter of **c** 30 nm and **d** 40 nm. Yellow lines emphasize crystal planes.

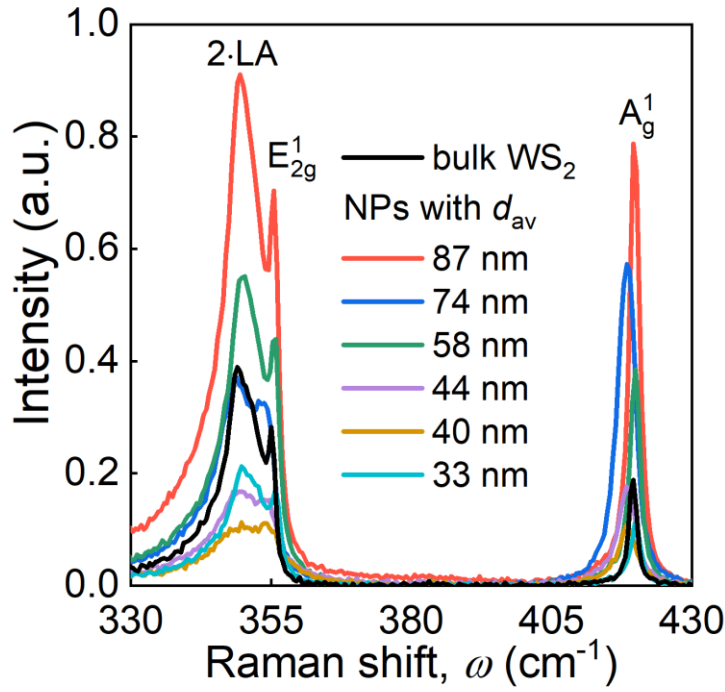
Supplementary Note 3. Selected area electron diffraction (SAED) analysis

SAED measurements have provided the ring sequence typical for polycrystalline MoS₂ and WS₂ with a hexagonal lattice (Figure 2b and Supplementary Figure 1b). The results of the analysis of these patterns are collected in Supplementary Table 1.

Supplementary Table 1. Values of interplanar spacings d_{exp} , determined from SAED pattern of MoS₂ and WS₂ NPs selected at 1000 rpm. d_{hkl} is a distance between consecutive crystallographic planes with Miller indices (hkl).

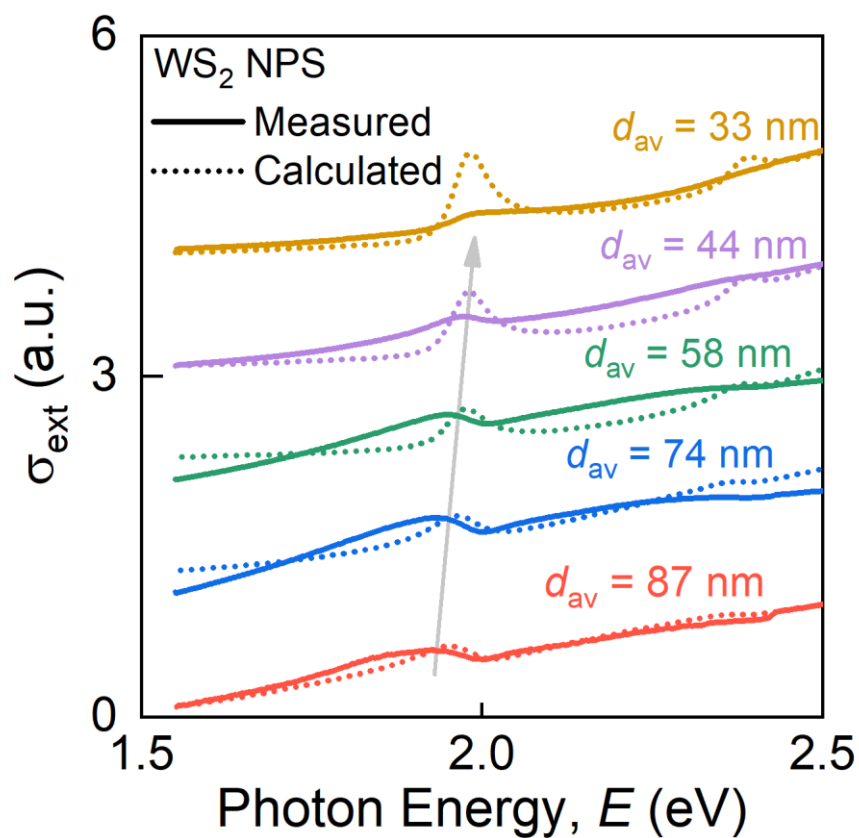
MoS ₂				WS ₂			
Ring	d_{exp} (Å)	hkl	d_{hkl} (Å)	Ring	d_{exp} (Å)	hkl	d_{hkl} (Å)
1	2.71	100	2.76	1	6.12	002	6.16
2	2.28	006	2.33	2	2.65	011	2.66
3	1.8	111	1.69	3	2.42	006	2.37
4	1.56	110	1.59	4	1.51	111	1.56
5	1.35	200	1.38	5	1.32	200	1.36

Supplementary Note 4. Raman scattering of WS₂ NPs



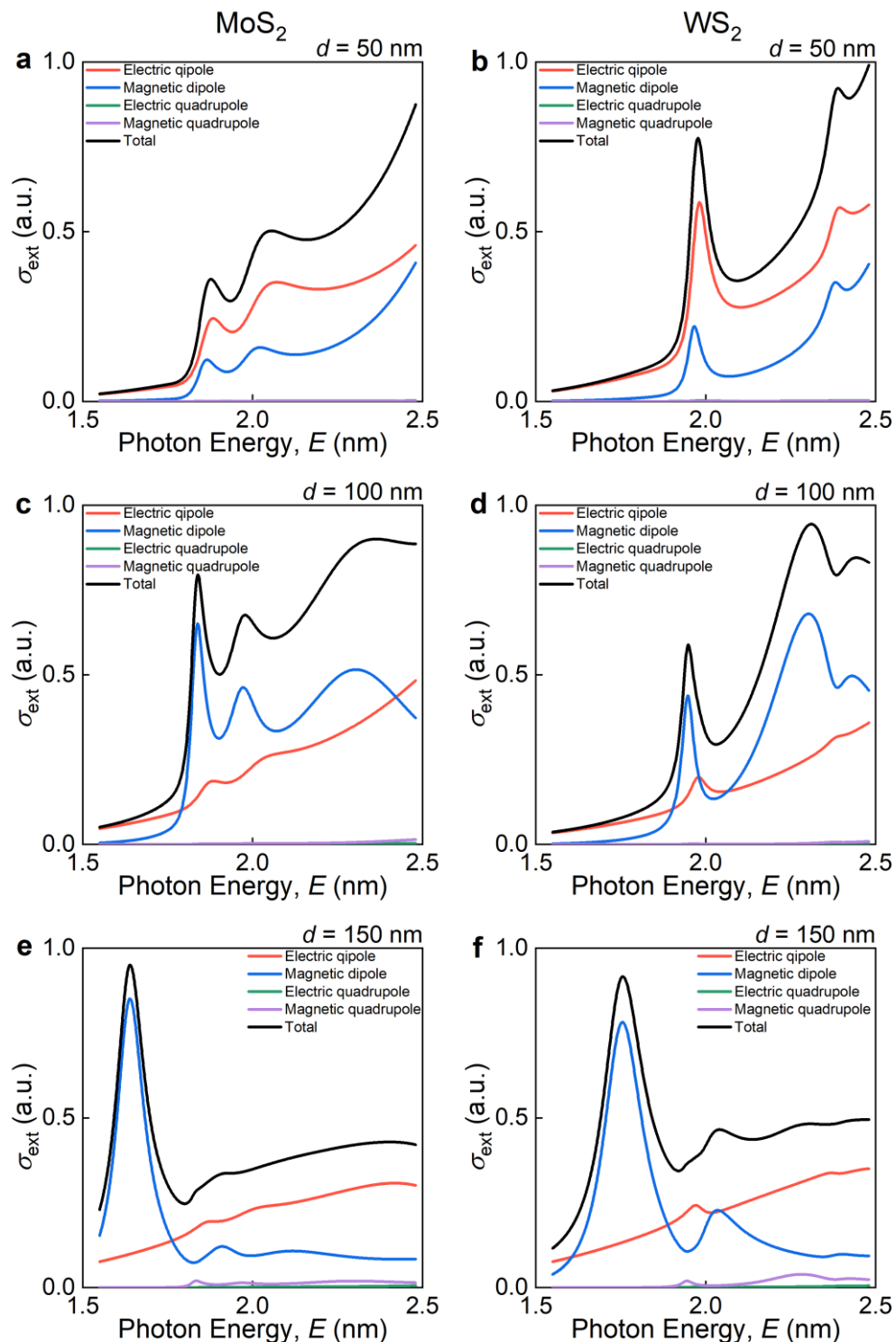
Supplementary Figure 3. Resonant Raman scattering spectra of laser-ablated WS₂ NPs separated by centrifugation at different rotation speed: 250, 500, 1000, 2000, and 4000 rpm, corresponding to d_{av} = 87, 74, 58, 44, 40, and 33 nm, respectively. For comparison, we included Raman scattering spectra of original bulk WS₂ crystal with hexagonal 2H symmetry. The excitation wavelength is 532 nm.

Supplementary Note 5. Extinction spectra of WS₂ NPs



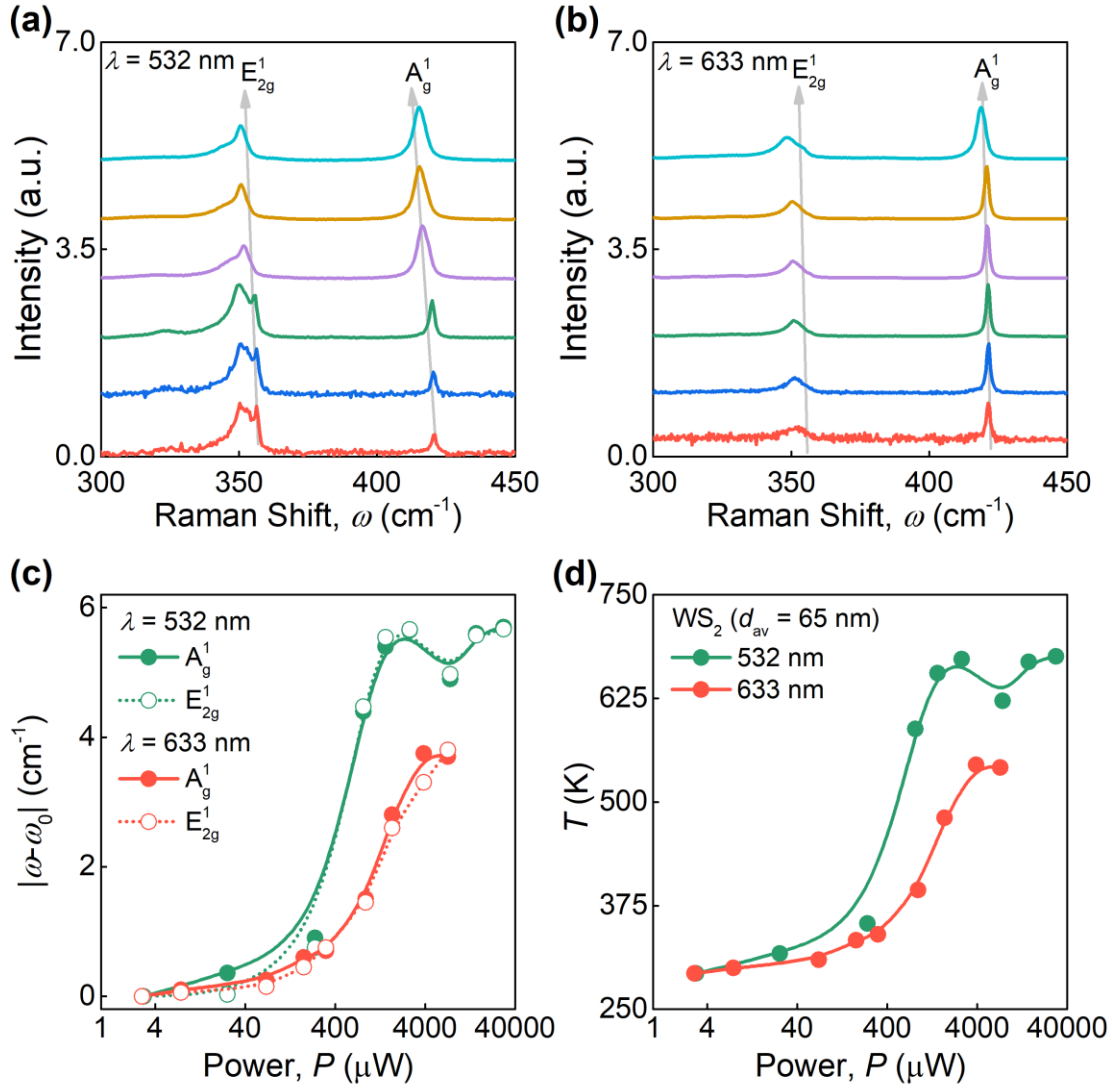
Supplementary Figure 4. Extinction cross-section of colloidal WS₂ NPs selected at the different rotation speeds of the centrifuge 250, 500, 1000, 2000, and 4000 rpm, corresponding to $d_{\text{av}} = 87, 74, 58, 44, 40$, and 33 nm, respectively. Solid and dashed lines show experimental and calculated data correspondingly.

Supplementary Note 6: Multipole analysis of TMDC NPs

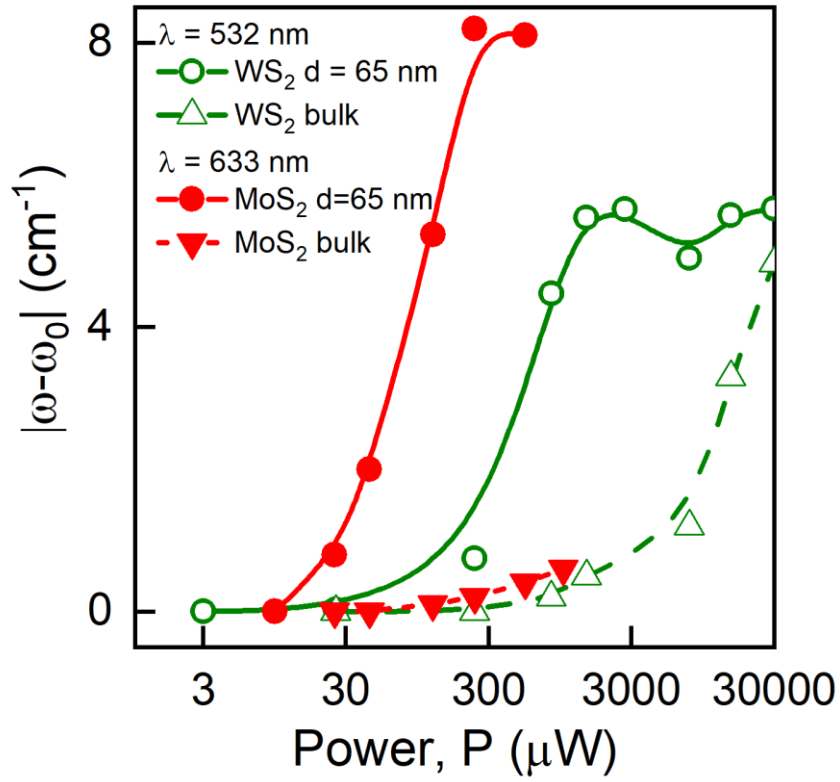


Supplementary Figure 5. Multipole decomposition of extinction cross-section of TMDC NPs. For **a-b** 50 nm, **c-d** 100 nm, and **e-f** 150 nm diameters.

Supplementary Note 7. Photothermal applications of WS₂ NPs



Supplementary Figure 6. Raman and photothermal responses of WS₂ NPs. Raman spectra of WS₂ NPs with an average size of 65 nm on glass substrate were recorded under different laser excitation power and at two laser wavelengths: **a** 532 nm and **b** 633 nm correspondingly. **c** Temperature shift of the frequencies corresponding to E_{2g}^1 and A_{1g} Raman scattering modes of WS₂ NPs with an average size of 65 nm on a glass substrate for two laser wavelengths $\lambda_{exc} = 532$ nm and $\lambda_{exc} = 633$ nm. **d** Dependence of WS₂ NPs local temperature on the power of exciting radiation for $\lambda_{exc} = 532$ nm and $\lambda_{exc} = 633$ nm.



Supplementary Figure 7. Temperature shift of the frequencies corresponding to E_{2g}^1 and A_{1g} Raman scattering modes of WS₂ and MoS₂ NPs with an average size of 65 nm in comparison with unpatterned crystals on a glass substrate for two laser wavelengths $\lambda_{\text{exc}} = 532$ nm and $\lambda_{\text{exc}} = 633$ nm.